

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and maintained **Python** applications and **data pipelines** processing financial transaction datasets at scale, using **Pandas** and **NumPy** for data analysis, transformation, and validation across 50+ daily batch jobs with **99.5% uptime SLA**.
- Implemented **PySpark**-based distributed data processing workflows for large-scale payments and settlements data, reducing pipeline execution time by **40%** and enabling real-time analytics for cross-functional teams including data engineers, QA, and DevOps.
- Wrote clean, efficient, and well-documented **Python** code following enterprise coding standards, participated in code reviews, and debugged complex issues in batch and data-processing jobs, achieving **95% code quality score** across 20+ production pipelines.
- Built and optimized **REST API** integrations using **Python Flask/FastAPI** to connect data pipelines with ERP and business systems, supporting **5+ downstream applications** with payment flows, transaction reconciliation, and regulatory compliance reporting.
- Demonstrated problem-solving and decision-making skills under pressure, managing **P1/P2 incidents** in production data systems, resolving **25+ critical data pipeline failures** within SLA windows and handling unexpected changes in data schema and processing requirements.
- Collaborated with banking-domain stakeholders and data teams to gather requirements, design data transformation logic for **ISO 20022** message processing, and deliver UAT-ready Python modules supporting **SEPA instant payments** and cross-border payment reconciliation workflows.

SKILLS

Python & Data Engineering – Python, PySpark, Pandas, NumPy, data pipelines, ETL, batch processing, data transformation, data validation.

Applications Development – REST APIs, Flask, FastAPI, Spring Boot, clean code practices, code reviews, performance optimization, debugging.

Databases & SQL – PostgreSQL, MySQL, Oracle, SQL query optimization, schema design, data warehousing.

Financial Services & Domain – ISO 20022 message flows, SEPA instant payments, cross-border payments, payment reconciliation, transaction processing, regulatory compliance, GDPR, change-control management.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins CI/CD pipelines, Git, Linux, monitoring and alerting.

Tools & Platforms – Jira, Confluence, Power BI, Tableau, R, scikit-learn, machine learning fundamentals.

Soft Skills – Cross-functional collaboration, written and verbal communication, problem-solving, decision-making, incident management, UAT support.

EDUCATION AND TRAINING

**MASTER OF SCIENCE in Data Analytics – National College of
Ireland, Dublin**

02/2026

PROJECTS

Large-Scale Payments Data Pipeline & PySpark Processing Engine (HCL Technologies)

- architected an end-to-end **Python and PySpark** data pipeline ingesting **10M+ daily payment transaction records** from SWIFT messaging and ERP systems, transforming **ISO 20022** message formats into normalized analytical datasets for **SEPA settlement** and cross-border payment reconciliation.
- Developed **Pandas**-based data validation and cleansing modules achieving **99.8% data accuracy**, identifying and remediating **50+ data quality issues** monthly through automated anomaly detection and flagging mechanisms integrated into the processing pipeline.
- Optimized **PySpark** job performance by implementing columnar partitioning, caching strategies, and join optimization, reducing batch execution time from **4 hours to 1.2 hours** and enabling near-real-time analytics for regulatory reporting and transaction visibility dashboards.
- Wrote production-grade, well-documented **Python** code adhering to enterprise standards, enabled seamless **REST API** integrations with downstream ERP and business systems, and conducted code reviews for 6+ junior developers, improving code quality score to **96%**.
- Automated debugging and monitoring of **batch jobs** using **CloudWatch** and custom **Python logging**, resolving **30+ data pipeline incidents** annually with average MTTR of **15 minutes**, and coordinating with DevOps and QA teams to implement preventive measures and post-incident reviews.

Banking Domain Payments Operations & Settlement Analytics Platform (HCL Technologies)

- Supported production monitoring and incident response for a **SWIFT messaging** dashboard and payment operations platform serving a global banking client, managing **P1/P2 incidents** with **99.7% SLA compliance** across transaction-flow visibility and operational alerting for **2M+ daily messages**.
- Developed **Python** automation scripts and SQL queries to validate payment transaction states, reconciliation workflows, and **ISO 20022** message compliance, enabling UAT teams and banking-domain business analysts to verify **PSD2 Open Banking** requirements and regulatory audit trails with **100% traceability**.
- Collaborated with banking operations teams and domain experts to implement change-control procedures for **SEPA instant payments** processing, schema updates, and **GDPR** data protection measures, supporting **8+ major system releases** with zero compliance violations.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)